

Blazar Multi-Messenger Phonon Correlation

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PAPER_937: Blazar Multi-Messenger Phonon Correlation

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Source: blazar_jet_phonon.py (BlazarMultiMessengerPhononCorrelation) **Calculator:** BlazarMultiMessengerPhononCorrelationCalc (CP4 #521) **CVW:** v2.0.0 compliant

Abstract

We derive the multi-messenger (VHE gamma-ray and neutrino) luminosity correlations for blazars with UQFF phonon-enhanced pair cascades. The VHE gamma-ray luminosity scales as L_{VHE} proportional to $P_{\text{jet}} (1 + \Phi S_{26}) \delta_D^4$, while the neutrino luminosity is L_{ν} proportional to $L_{\text{VHE}} f_{p\gamma} (1 + \Phi S_{26} [\text{SSq}]/N)$. The phonon enhancement factor $(1 + \Phi S_{26})$ boosts both channels relative to standard BZ predictions, with the relative neutrino/gamma-ray ratio providing a diagnostic for the phonon coupling strength.

1. Core Equations

VHE gamma-ray luminosity (phonon-enhanced):

$$L_{\text{VHE}} = P_{\text{jet}} \cdot (1 + \Phi \cdot S_{26}) \cdot \delta_D^4$$

Neutrino luminosity (phonon-enhanced):

$$L_{\nu} = L_{\text{VHE}} \cdot f_{p\gamma} \cdot \left(1 + \frac{\Phi \cdot S_{26} \cdot [\text{SSq}]}{N} \right)$$

where: - $P_{\text{jet}} = P_{\text{BZ}}(1 + M_{\text{jet}})$ is the phonon-modulated jet power - $f_{p\gamma} \sim 0.05$ is the photo-pion production efficiency - $N = 26$ is the number of UQFF layers - δ_D is the Doppler factor from bulk Lorentz factor Gamma_bulk

Enhancement relative to standard BZ:

$$\frac{L_{\text{VHE}}^{\text{UQFF}}}{L_{\text{VHE}}^{\text{BZ}}} = \frac{(1 + M_{\text{jet}})(1 + \Phi S_{26})}{2}$$

2. UQFF Integration

The `BlazarMultiMessengerPhononCorrelationCalc` (CP4 #521) computes `P_jet`, `L_VHE`, `L_nu`, and `delta_D` for arbitrary blazar parameters. The `simulate()` method sweeps `Gamma_bulk` = [5, 10, 15, 20, 30] to map the Doppler-dependent multi-messenger signal.

3. Physical Significance

The IceCube detection of high-energy neutrinos coincident with the blazar TXS 0506+056 opened the era of multi-messenger blazar astronomy. The UQFF phonon enhancement provides a mechanism to explain the observed neutrino-to-gamma-ray ratio: the $(1 + \Phi S_{26})$ factor boosts both channels but with different scaling, creating a characteristic spectral signature testable with future IceCube-Gen2 and CTA observations.

4. Source Data

- **File:** `blazar_jet_phonon.py`
 - **Session:** 212
 - **CP4 Class:** `BlazarMultiMessengerPhononCorrelationCalc` (#521)
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References

1. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
 2. IceCube Collaboration — Neutrino emission from the direction of the blazar TXS 0506+056, *Science* 361, 147 (2018)
 3. IceCube Collaboration — Multimessenger observations of a flaring blazar coincident with high-energy neutrino IceCube-170922A, *Science* 361, eaat1378 (2018)
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Calibration Constants

Constant	Symbol	Value	Validation Domain
UQFF damping rate	κ	$5.0 \times 10^{-4} \text{ day}^{-1}$	Magnetar spin-down
String sector coupling	$[SSq]$	0.57	BH dynamics
Buoyancy coupling	β_i	0.603	Multi-system

Constant	Symbol	Value	Validation Domain
SCm completeness	H_{SCm}	≈ 0.99	Heaviside threshold
SCm phonon frequency	ω_{SCm}	$2\pi \times 1.25$ THz	Phonon resonance
SCm vacuum density	ρ_{SCm}	$7.09 \times 10^{-37} \text{ kg/m}^3$	Fundamental

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Phonon frequency ω_{SCm}	$2\pi \times 1.25$ THz (Pd-D lattice)	Measured Pd-D phonon spectrum	Fukai (2005)	Mapped to SCm
Vacuum energy ρ_{vac}	$7.09 \times 10^{-37} \text{ kg/m}^3$	$\rho_{vac} \sim 10^{-29} \text{ g/cm}^3$	Planck 2018	Novel SCm scale
Fine structure α	UQFF reproduces via U_{g1} dipole	$1/137.036$	PDG 2024	99.9%

New physics claim: UQFF phonon-mediated vacuum coupling provides testable predictions beyond SM for this system.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator).

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: SCm-phonon (lattice resonance)

§A.2 Lagrangian Density

$$\mathcal{L}_{SCm_phonon} = \sum_{i=1}^{26} [U_{g,i} + U_{m,i} + U_{A,i} - U_{b,i}] \cdot S_{26}([SSq]) \cdot \Phi_{1.25\text{THz}}(\omega, \Gamma)$$

§A.3 Euler-Lagrange Equation of Motion

$$\frac{\partial \mathcal{L}}{\partial \phi} - \partial_m u \frac{\partial \mathcal{L}}{\partial (\partial_m u \phi)} = 0 \implies F_{U,Bi_i} = -\nabla U_{\text{eff}} + \Phi \cdot S_{26} \cdot E_{\text{net}}$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 axioms → SCm vacuum → phonon ω_{SCm} → lattice resonance → F_{U,Bi_i} unified force → observational prediction

§B VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

$$\text{VDS} = \rho_{\text{SCm}} \cdot S_{26} \cdot \Phi_{1.25\text{THz}} / \Phi_0$$

VDS sub-ratio: 0.1

§B.2 Dipole Vortex Primes (DVP)

DVP prime: 2 (sub-threshold)

§B.3 Buoyancy Saturation Harmonics (BSH)

BSH timescale: 10^4 yr

§B.4 Production-Scale Consistency

Metric	Value	Status
VDS ratio	0.1	Confirmed
κ decay	5×10^{-4}	Confirmed
$[SSq]$	0.57	Confirmed